

Perceptron learning and Hoeffding inequality

1. Simulate the Perceptron Learning Algorithm

Consider the following target function:

x_1	x_2	$y = f(x)$
-1	-1	-1
-1	1	-1
1	-1	1
1	1	1

Complete the missing rows in the table below, training a 2D perceptron for 8 steps ($t = 0..7$) to learn the target function, using the update rule of the Perceptron Learning Algorithm.

t	x_1	x_2	w_0	w_1	w_2	$\hat{y}$	correct?	w'_0	w'_1	w'_2
0	-1	-1	0	0	0	0	×	1	1	-1
1	-1	1	1	1	-1	-1	✓			
2	1	-1								
3	1	1								
4	-1	-1								
5	-1	1								
6	1	-1								
7	1	1								

Does the algorithm converge in 8 steps? Is the dataset linearly separable?

2. Hoeffding inequality

You are testing a hypothesis on 43 samples, sampled from the data distribution. After measurement, you find your error to be 0.1. Your boss told you that she will tolerate at most an error of 0.2 in production (averaged for the long run). Derive an upper bound for the probability that the error will be larger than that (and she will be unhappy with you)?

3. Implement the Perceptron Learning Algorithm

The inputs for the tasks are in Numpy's npz format. They contain 4 arrays: X, Y for the training inputs and outputs, and X_test and Y_test for the test inputs and outputs, respectively. The samples are the rows of the X matrices.

Use only the training data for calculating the parameters (weights). Some input files contain reference solutions, you can use these to check the correctness of your answer (but only for that).

- a. Find a linear separator that separates the two classes (-1 and 1) in the `pla.npz` dataset.
- b. Plot the training data and the separator in each iteration. (You can use the code in `visualise.py`.)¹

A simple way to do nonblocking plot in Matplotlib is:

```
import matplotlib.pyplot as plt
from visualise import visualise
fig, ax = None, None
while not done:
    w = calculate()
    fig, ax = visualise(w, X, Y, block=False, fig=fig, ax=ax)
    plt.pause(0.5)
```

- c. Repeat a) and b) on `pocket.npz` as well. How did it perform? Can you do better with a small modification?

¹To plot an animation, you can use the guides here: https://matplotlib.org/stable/api/animation_api.html